

Patent–Research Classification System and Workflow Specification

This document provides a complete, high-level specification for: - Building a **controlled vocabulary** from research papers and patents - Tagging each record with extracted vocabulary - Applying a unified **classification system** to both datasets - Preparing the resulting data for gap analysis

This document is intended for use by human analysts and AI agents performing the classification.

1. Project Goal

To create a unified, structured representation of **4,000+ research papers** and **all collected patents** related to ferrofluids so we can:

1. **Identify research → application gaps**
2. Detect **under-patented scientific fields**
3. Find **high-potential, low-competition invention spaces**
4. Map relationships between fundamental science and commercialized technologies

This requires: - A shared **classification system** for both research and patents - A domain-specific **controlled vocabulary** extracted from both corpora - A standardized AI-assisted workflow

2. Overview of the Complete Workflow

The full process follows these stages:

Phase 1 — Controlled Vocabulary Generation

1. Extract text from all papers and patents (title, abstract/summary, keyword fields).
2. Perform n-gram and phrase extraction (unigrams, bigrams, trigrams).
3. Remove generic stopwords and keep domain-specific terms.
4. Cluster terms into conceptual groups (e.g., heat transfer, droplet physics, hyperthermia, CFD, bearings).
5. Merge vocabulary from research and patents into a single **controlled vocabulary**.
6. Assign vocabulary clusters semantic labels.

Output: A curated vocabulary list with domain groupings.

Phase 2 — Tagging All Records Using the Controlled Vocabulary

1. For each paper and patent, scan the text.
2. Identify all controlled vocabulary terms.
3. Attach the detected terms to the record as **tags**.
4. Tag frequency and cluster membership reinforce classification decisions.

Output: Each record now has its own list of domain-relevant tags.

Phase 3 — Unified Classification Using the Tag-Enriched Records

For **both** research papers and patents: 1. Determine **Doc_Type** (Article, Review, BookChapter, Patent, Other). 2. Identify the **main domain** (Materials, Computation, Experimentation, Applications). 3. Assign **Primary_Class** (one of 11–49). 4. Optionally assign **Secondary_Class** and **Tertiary_Class**. 5. Generate short **classification reasoning**.

For patents only: - Assign an **Innovation_Type**: New_Material, New_Device, New_Method, New_Use, New_Process, New_System.

Output: A structured classification for each of the ~4,000 records and all patents.

3. Unified Classification System (v2)

This taxonomy is used **identically** for research papers and patents. Patents receive an additional `Innovation_Type` tag.

3.1 Doc_Type (All Records)

- Article
 - Review
 - BookChapter
 - Patent
 - Other
-

3.2 Content Classes (11–49)

MATERIALS (11–16)

11 – Material / Chemistry – Chemical synthesis, precursors, doping. **12 – Material / Formulation** – Carrier fluids, surfactants, stabilizers. **13 – Material / Properties** – Viscosity, magnetization, rheology. **14 –**

Material / Evaluation & Characterization – VSM, XRD, TEM, SEM, DLS. **15 – Material / Manipulation & Droplets** – Droplet deformation, breakup. **16 – Material / Handling** – Storage, sedimentation, safety.

COMPUTATION & MODELLING (21–29)

21 – Finite Element Analysis (FEA) – FEM, COMSOL, ANSYS. **22 – CFD** – Navier–Stokes, FLUENT, OpenFOAM. **23 – MATLAB-based Studies** – MATLAB/Simulink algorithms. **24 – Analytical Models** – ODEs, BVPs, dipole models, susceptibility. **25 – Flow Property Modelling** – Pressure drop, velocity fields. **26 – Heat Transfer Modelling** – Thermal conductivity, Nusselt numbers. **27 – Other Modelling** – Optimization, neural networks. **28 – Stability Modelling** – Sedimentation, aggregation models. **29 – Magnetic Droplet Modelling** – Spin dynamics, shape analysis.

EXPERIMENTATION & INSTRUMENTATION (31, 33)

31 – Experimental Systems – Rigs, prototypes, device testing. **33 – Instrumentation** – Sensors and instruments using ferrofluids.

APPLICATIONS (30, 32, 38–49)

30 – Other Applications – Applications not elsewhere classified. **32 – Linear Motors & Encoders**. **38 – Magnetic Induction Devices**. **39 – Medical – Robotic Surgery**. **40 – Medical – Hyperthermia, Cancer, Drug Delivery**. **41 – Medical – Pharmaceutical**. **42 – Biomedical (Non-Pharmaceutical)**. **43 – Robotics – General**. **44 – Geology & Oil-field Recovery**. **45 – Flow-based Devices**. **46 – Heat Transfer Devices**. **47 – Bearings, Seals, Lubricants**. **48 – Levitation & Droplet Spin**. **49 – Digital Microfluids, Damping, Environmental**.

3.3 Patent-Only: Innovation_Type

- New_Material
- New_Device
- New_Method
- New_Use
- New_Process
- New_System

4. Agent Workflow Instructions

These instructions are for the AI agent performing the classification.

Step 1 — Determine Doc_Type

Use metadata or text cues: - "review", "survey" → Review - "book chapter" → BookChapter - Patent dataset → Patent - Otherwise → Article

Step 2 — Use Controlled Vocabulary to Identify Domains

Review all detected tags and determine which domain (Materials / Computation / Experimentation / Applications) the paper or patent primarily belongs to.

Step 3 — Assign Primary_Class

Choose **one** class (11–49) that represents the **main scientific or inventive contribution**. For patents: focus on the **core inventive concept**, not background science.

Step 4 — Secondary and Tertiary Classes (Optional)

Add only when the paper/patent clearly spans distinct domains.

Step 5 — Patent-Only: Assign Innovation_Type

Identify what form of innovation is being claimed.

Step 6 — Provide Classification Reasoning

Give a short explanation referencing: - Key phrases in the abstract - Controlled vocabulary tags - Features of the invention or scientific contribution

5. Final Data Schema

Each record (paper or patent) should output: - DOI / Patent Number - Year - Title - Doc_Type - Controlled_Vocab_Tags - Primary_Class - Primary_Desc - Secondary_Class (optional) - Tertiary_Class (optional) - Innovation_Type (patents only) - Reasoning

6. Intended Analytical Uses

Once all records are classified: 1. Compare class frequencies between **research** vs **patents**. 2. Identify research domains with **low patent activity** → innovation gap. 3. Identify patent-heavy domains with **weak scientific underpinnings** → opportunity for deeper research. 4. Map controlled vocabulary clusters to find **scientific concepts not yet applied**. 5. Run network/linkage analysis between vocabulary clusters and application classes.

This completes the high-level system specification for the patent–research classification pipeline.